

Does spectral information beyond RGB improve land-use classification? A controlled study on EuroSAT

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Summary. We compare an RGB baseline against several multispectral approaches for land-use classification on Sentinel-2 imagery. Rather than optimise a single number on a saturated benchmark, we treat the RGB-versus-multispectral question as a controlled experiment. We find that spectral information helps *conditionally*: strongly for classical features (+12 pp) and for in-domain-pretrained or from-scratch networks, but negligibly—or harmfully—for ImageNet-pretrained networks, especially with few labels. The value of extra bands depends on the representation that consumes them.

1 Problem formulation

Land-use and land-cover classification from satellite imagery underpins applications from agricultural monitoring to infrastructure mapping. Beyond the visible bands, Sentinel-2 records red-edge, near-infrared (NIR) and short-wave infrared (SWIR) channels that encode surface properties invisible to the eye. We study EuroSAT [1], ten land-use classes over 27,000 patches, and ask: does information beyond RGB improve classification, and *where and why*? Because the benchmark is near-saturated, a single accuracy figure is uninformative; the scientific content lies in isolating the contribution of spectral information from confounds such as the pretraining domain and the label budget. We therefore build an RGB baseline and several multispectral approaches, evaluate them under one identical 80/20 split, and analyse the conditions under which extra bands help.

2 Data and preprocessing

EuroSAT provides 27,000 georeferenced patches, 64×64 px at 10 m, in ten classes and 13 Sentinel-2 bands stored as Level-1C top-of-atmosphere (TOA) reflectance scaled by 10^4 [1, 2]. We use the all-band GeoTIFFs and build the RGB baseline from bands B4/B3/B2 of the *same* files, avoiding the compression confound of the separate lossy JPEG archive. Two data facts drive preprocessing (Fig. 1). First, the GeoTIFFs store B8A as the *last* channel rather than in canonical order—a silent index footgun, guarded here by a single named band map with unit tests. Second, band B10 (cirrus) is near-empty over these cloud-screened patches (train mean 12.1 DN versus 946–2299 for surface bands), so we exclude it and label the multispectral arm “12-band”.

We draw a stratified 80/20 train/test split with a committed manifest and hold out 10% of the training portion for validation (19,440/2,160/5,400 patches); the split is identical for every arm, which is what makes the paired significance test valid. Reflectance channels are z -scored using train-split statistics; the in-domain-pretrained arms instead divide by 10^4 to match their pretraining distribution, so normalisation is a per-arm property. We derive four spectral indices—NDVI, NDWI [10], NDBI and the red-edge NDRE—from raw DN (normalised differences are scale-invariant), ε -guarded and clipped to $[-1, 1]$. Augmentation is restricted to flips and 90° rotations: photometric jitter would corrupt the physical meaning of reflectance and the indices derived from it.

3 Methods

The system has two tiers (Fig. 2). **Tier 1 (classical)** trains a Random Forest on per-band and per-index summary statistics; it is interpretable and yields permutation importance directly. Arm C1 uses RGB statistics; C2 uses the 12 bands plus the four indices.

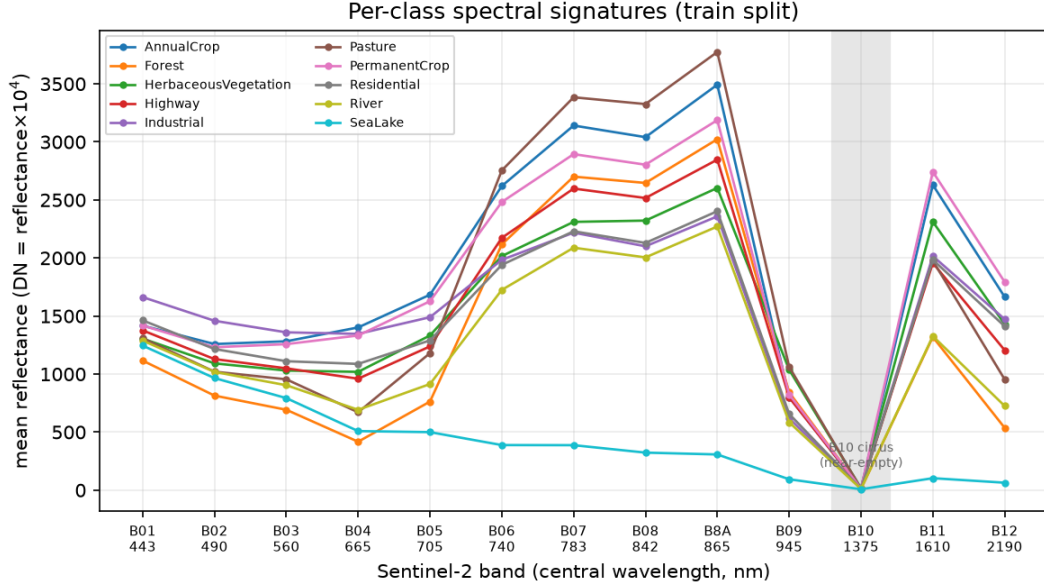


Figure 1: Per-class mean spectral signatures. Classes separate most in NIR/SWIR, not RGB; the highlighted cirrus band B10 is near-empty, justifying its removal.

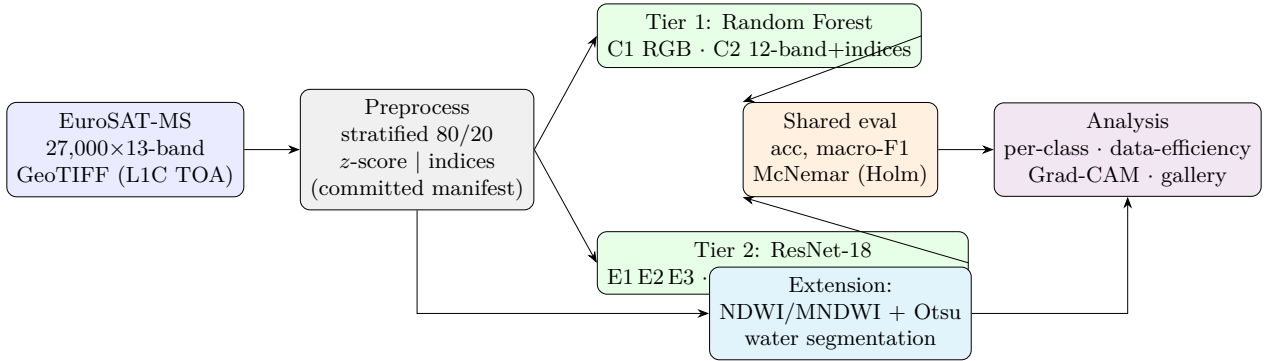


Figure 2: System pipeline. One preprocessing path and one shared evaluator feed every arm, so the classical and deep tiers are compared on an identical split.

Tier 2 (deep) fine-tunes a ResNet-18 [3] under one frozen recipe shared across all arms (AdamW, cosine schedule, ≤ 30 epochs, early stopping on validation macro-F1), so arms differ only in inputs and pretraining, never in tuning. Arm **E1** is the ImageNet-pretrained RGB baseline; **E2** ingests the 12 bands by inflating the first convolution—timm replicates the pretrained RGB filters across channels and rescales by $3/N$ to preserve activation magnitude [4, 5]; **E3** appends the four indices to RGB. To separate *pretraining domain* from *spectral information*, **E5a/E5b** repeat RGB and 12-band from scratch (no pretraining), and **E7/E7b** use SSL4EO-S12 weights—a ResNet-18 self-supervised on Sentinel-2 [6, 7]—for the 13-band and RGB inputs. E1/E2 and E7b/E7 together form a $\{\text{ImageNet, SSL4EO}\} \times \{\text{RGB, all-band}\}$ design (Fig. 4). As an optional extension we segment water by thresholding NDWI/MNDWI [11] with a fixed cut and with Otsu’s method [12].

4 Experimental setup

We follow the mandated 80/20 protocol; the validation slice is used for model selection and the test set is scored once. The arms whose results carry the argument—E1, E2, E3 and the E6 data-efficiency sweep at 1/5/10/25% of the labels—run at three seeds (reported as mean $\pm$ std with per-seed paired tests); the supporting confound-control and SSL4EO arms run at a single seed (point estimates) under a fixed compute budget. We report accuracy (mandated) alongside macro-F1 and

Arm	Input / pretraining	Accuracy (%)	Macro-F1 (%)
<i>Deep, ImageNet-pretrained</i>			
E3	RGB + indices	98.69 $\pm$ 0.07	98.65
E1	RGB (baseline)	98.41 $\pm$ 0.12	98.36
E2	12-band	98.33 $\pm$ 0.40	98.27
<i>Deep, from scratch / SSL4EO (1 seed)</i>			
E5b	12-band, scratch	98.17	98.08
E5a	RGB, scratch	96.54	96.41
E7	13-band, SSL4EO	97.17	97.05
E7b	RGB, SSL4EO	94.37	94.25
<i>Classical (Random Forest, 1 seed)</i>			
C2	12-band + indices	91.44	91.09
C1	RGB stats	79.28	78.47

Table 1: Full-data test accuracy (5,400-patch test set, identical split). ImageNet arms tie; spectral information helps under SSL4EO, from scratch, and classically.

Method	NDWI+fixed	NDWI+Otsu	MNDWI+fixed	MNDWI+Otsu
micro-IoU	0.443	0.509	0.354	0.441
micro-Dice	0.614	0.675	0.523	0.612

Table 2: Water segmentation on 11 hand-labelled River/Sea-Lake patches.

per-class recall, given the mild 1.5:1 class imbalance. Significance uses McNemar’s test [8] on the two arms’ identical test predictions, per matched seed; the comparison family is each multispectral arm versus the RGB baseline, Holm-corrected for multiplicity [9].

5 Results

Full-data accuracy (Table 1) clusters all deep arms within 0.4 pp at 98.3–98.7%; E3 (RGB+indices) is nominally best at 98.69 $\pm$ 0.07%. McNemar for E1 versus E2 yields 109–151 discordant test patches and reaches significance in only one of three seeds after correction: at full data RGB and the 12-band model are statistically indistinguishable—the expected ceiling effect. E3 edges E1 in one seed.

Data efficiency (Fig. 3) tells the opposite of the textbook hypothesis. With ImageNet pre-training, RGB *beats* multispectral when labels are scarce: at 1% of labels ($\sim$ 190 images) RGB scores 81.4 $\pm$ 0.8% against 74.7 $\pm$ 2.5% (−6.7 pp, and noisier); the gap shrinks monotonically to 93.3/91.9 at 5%, 95.7/95.3 at 10%, and vanishes by 25% (96.7/96.8).

Tier-1 reveals the largest spectral effect of the study. Multispectral features (C2, 91.4%) beat RGB features (C1, 79.3%) by +12.2 pp, with an overwhelmingly significant McNemar result ($p \approx 7 \times 10^{-106}$, 901 discordant). The per-class gains (Fig. 6) concentrate exactly where physics predicts: River +28 pp, Highway +24, Herbaceous Vegetation +23, Permanent Crop +20; the top permutation-importance features are the NDWI and NDRE indices.

The 2 $\times$ 2 (Table 1, lower block) is decisive: under SSL4EO Sentinel-2 pretraining the all-band model beats RGB (97.2 vs 94.4%, +2.8 pp)—the reverse of the ImageNet rows where the two tie. Training from scratch agrees: 12-band (E5b, 98.2%) beats RGB (E5a, 96.5%). The subjective gallery (Fig. 5) shows the same classes the confusion analysis flags—water and crop types—separating cleanly in false-colour and NDWI views that RGB lacks.

Segmentation (Table 2): on 11 hand-labelled patches, NDWI+Otsu is best (micro-IoU 0.509, Dice 0.675). Otsu beats a fixed threshold for both indices, and NDWI beats MNDWI—plausibly because the SWIR bands feeding MNDWI are upsampled from 20 m and blur the thin ($\sim$ 3 px) rivers.

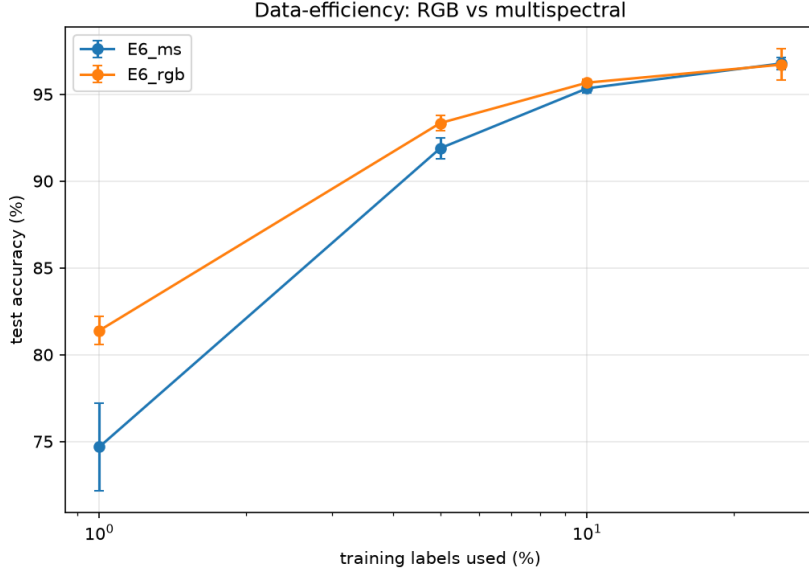


Figure 3: Data-efficiency curve. Under ImageNet pretraining, multispectral trails RGB most when labels are scarce; the gap closes as data grows.

	RGB	all-band
ImageNet	98.4	98.3
SSL4EO	94.4	97.2

Figure 4: Design 2×2 (test accuracy, %): extra bands help only when pretraining is in-domain (+2.8 pp, bottom row); under ImageNet they tie.

6 Discussion

The results resolve “does multispectral help?” into “it depends on the representation that consumes it.” Spectral information clearly helps in three regimes: (i) classical features with no deep pretraining (+12 pp); (ii) deep networks trained from scratch ($E5b > E5a$); and (iii) deep networks pretrained in-domain (SSL4EO all-band $>$ RGB). It does *not* help ImageNet-pretrained networks at full data—where they tie and McNemar is non-significant—and it actively *hurts* them at low labels, where a 12-channel stem must be adapted from RGB-shaped filters with too few examples. This reconciles our study with Helber et al., who reported RGB as their best band combination [1]: their networks were ImageNet-pretrained, so their finding is specific to that regime rather than a property of the data. The Tier-1 per-class gains tie back to band physics—water and impervious surfaces separate through NIR/SWIR and NDWI, crop types through the red edge—which is exactly the information an RGB-only model cannot see. For an applied multimodal setting, the practical reading is that additional channels pay off most when the backbone is pretrained in-domain or trained on the target data, and least when leaning on an RGB-pretrained representation, particularly under label scarcity.

7 Limitations and improvements

The mandated random 80/20 split ignores spatial autocorrelation: EuroSAT patches cropped from shared Sentinel-2 scenes can leak between train and test, inflating absolute accuracy [13]. The RGB-versus-multispectral *delta* shares the split and is partially insulated, but a spatial-holdout robustness

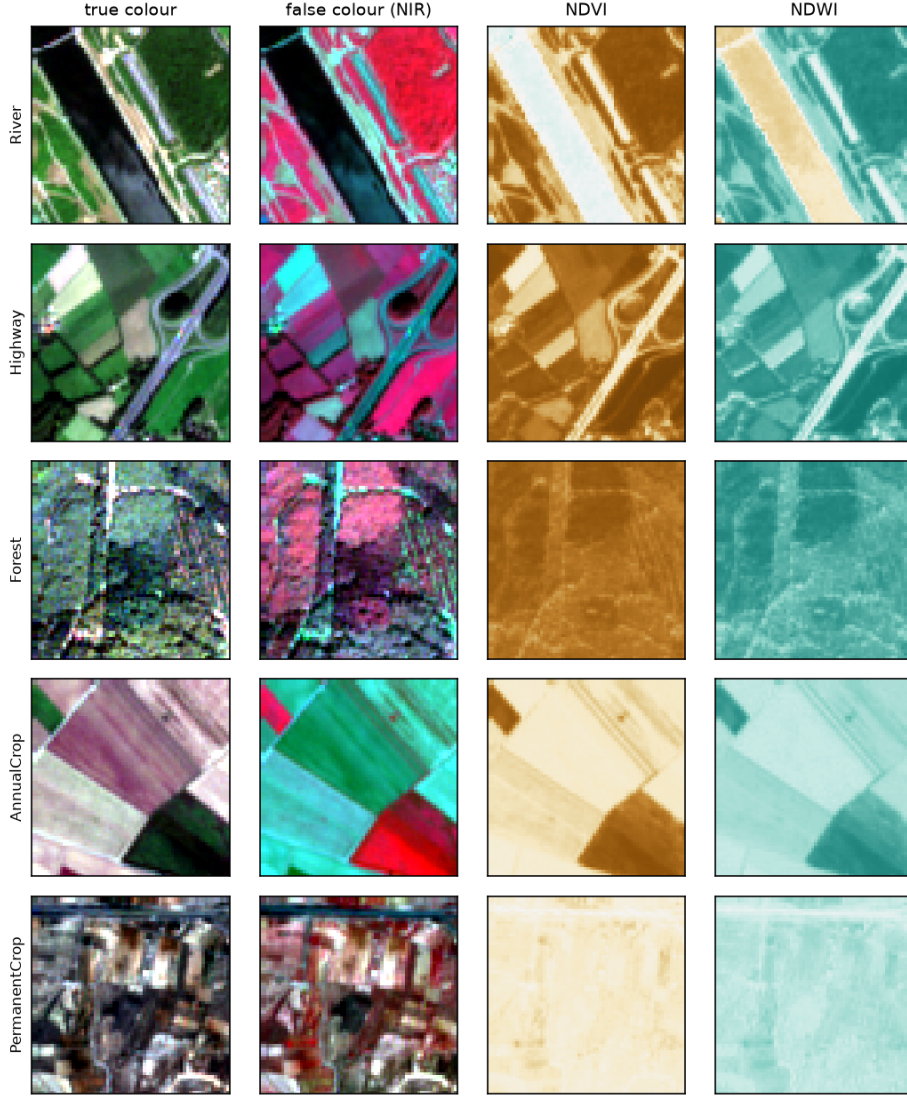


Figure 5: Subjective sample, one patch per class: true-colour, false-colour (NIR), NDVI and NDWI. Water and vegetation separate in the NIR/index views that RGB cannot see.

check remains future work. The data are L1C TOA without atmospheric correction, so indices are discriminative features, not calibrated physical quantities, and the 20/60 m bands are cubic-spline upsampled, leaving SWIR/red-edge effectively lower-resolution (visible in the segmentation result). With three seeds, the reported standard deviations are indicative rather than inferential—our significance claims rest on McNemar, not on overlapping error bars—and the supporting arms are single-seed point estimates. The segmentation evaluation uses only 11 patches with thin rivers, so its variance is high and it should be read as illustrative. Natural extensions are a spatial-holdout evaluation, larger or transformer backbones, L2A surface-reflectance inputs, more seeds, and semi-supervised methods for the low-label regime [14].

8 Reproducibility

The repository contains the full pipeline, committed split manifests, normalisation statistics, per-seed metrics and all figures. `scripts/download_data.py` fetches the checksum-verified dataset and `scripts/run_all.sh` reproduces every result; test suites guard band order, split integrity and label subsampling.

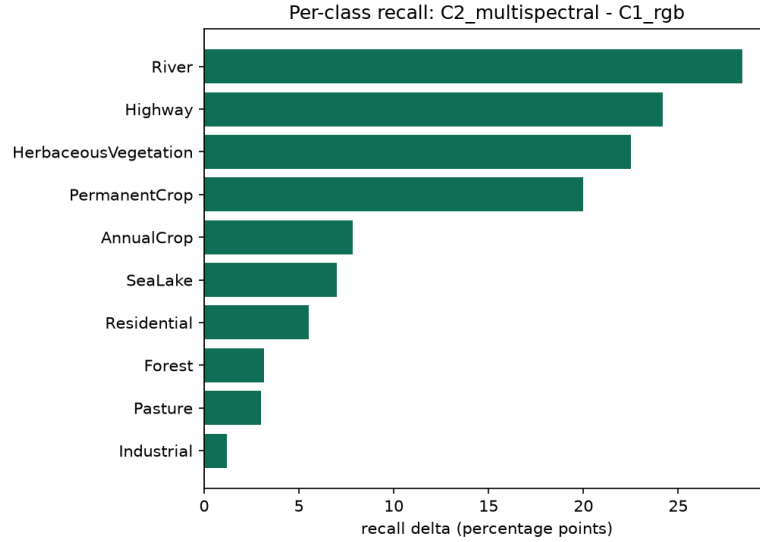


Figure 6: Tier-1 per-class recall gain from multispectral features (C2–C1). Gains concentrate on water/impervious and crop-type classes.

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